

Multi-Tenant Restaurant POS & Management SaaS Engine

A High-Performance Backend Ecosystem for Real-Time Dining, Multi-Station Order Flow, Dynamic Kitchen KDS, and Centralized Multi-Tenant Governance

PRIMARY FRAMEWORK
NestJS 11 (TypeScript)

DATA & ORM LAYER
PostgreSQL + Prisma
ORM

SECURITY & RBAC
6-Tier Hierarchical RBAC
+ JWT

INDUSTRY DOMAIN
Food & Beverage /
Restaurant SaaS

45% ⚡

Latency Reduction
Wait-to-kitchen order
dispatch

100% 🔒

Tenant Isolation
Zero cross-restaurant
data leak

6 Roles 👥

RBAC Hierarchy
Super Admin down to KDS
cook

Multi-MFS 💳

Payment Channels
Cash, Card, bKash,
Nagad, etc.

1 Executive Summary & Business Challenge

Modern food and beverage businesses operate in high-tempo, multi-station environments where milliseconds matter. Traditional single-station POS systems suffer from operational bottlenecks: waiter handwriting delays, kitchen communication chaos, inventory shrinkage, lack of real-time table turnover metrics, and fragmented payment reconciliation (combining cash, cards, and mobile financial services).

The Solution: This backend platform implements a high-throughput, cloud-native Multi-Tenant SaaS designed to empower independent restaurant owners (Supervisors/Managers) as well as global SaaS franchisors (Super Admins). With strict tenant data isolation, real-time KDS pipeline synchronization, visual table floor maps, instant SKU/barcode stock engines, and four-digit quick PIN authentication for high-speed counter handovers, the system achieves frictionless restaurant orchestration.

✖ Traditional Restaurant Bottlenecks

Before
SaaS

- Order Latency:** Waiters write paper slips, physically deliver them to kitchen, causing 10–15 min prep delay.
- Billing Discrepancies:** Manual discounts, voucher calculations, and untracked split payments lead to cash shortages.
- Stock Blindspots:** No real-time stock deduction on dish sales; frequent unexpected 86'd items during rush hour.

✔ SaaS Architecture Transformation

After
Solution

- Instant KDS Stream:** Floor orders dispatch in sub-100ms directly to kitchen display screens with modifiers.
- Omnichannel POS Hub:** Split settlement via Cash, Cards, and MFS (bKash, Nagad, Rocket, Upay) with automated voucher engine.
- Automated SKU Engine:** Integrated barcode/QR scanner, real-time inventory decrement, and low-

- **Multi-Tenancy Vulnerabilities:** Complex setups risk accidental data sharing between franchise branches.

stock alerts.

- **Strict Tenant Scoping:** Scoped database queries keyed by `businessId` guarantee 100% data partition.

USER ROLE	ACCESS SCOPE	KEY PERMITTED ACTIONS	RESTRICTED BOUNDARIES
SUPER_ADMIN	Global Platform	Tenant onboarding, SaaS plan pricing, platform telemetry	Does not alter daily kitchen tickets
SUPERVISOR	Tenant Wide	Staff approval gates, live sales overview, discount rules, KDS oversight	Cannot inspect other tenant databases
MANAGER	Branch Floor	Table configurations, stock adjustment, menu price updates	Cannot modify global subscription plans
CASHIER	Billing Station	Order settlement, split payments (Cash/MFS), receipt generation	Cannot approve new staff accounts
SERVER / WAITER	Dining Floor	Table status toggle, instant order dispatch with dietary tags	Cannot complete financial checkout
KITCHEN	KDS Line	Bump tickets (PREPARING → READY), view cook modifiers	No access to billing or revenue totals

4 Core Functional Modules & API Architecture

The backend is split into 14 decoupled feature modules. Each endpoint enforces strict data contracts, validation pipes, and Swagger OpenAPI schemas.

FUNCTIONAL MODULE	ROUTE PREFIX	METHOD	TECHNICAL CAPABILITY & LIFECYCLE
Executive Analytics	/overview	GET	Calculates live daily revenue, active terminal heartbeats, transaction counts, and pending order backlogs.
Staff & Approvals	/users/:id/approval	PATCH	Supervisor security gate: accepts or blocks employee accounts; maintains masked PINs for high-speed terminal login.
Order Engine	/orders	POST GET	Orchestrates complete order state machine: PENDING → PREPARING → READY → SERVED → COMPLETED.
Cashier POS Hub	/cashier/checkout	POST	Multi-channel settlement engine supporting Cash, Card, and MFS (bKash, Nagad, Rocket, Upay) with voucher validation.
Kitchen KDS Stream	/kitchen/tickets/:id/bump	PATCH	Real-time ticket bump stream for cook stations with prep time tracking and station capacity analytics.
Floor Map & Waiter	/serve/tables	GET PATCH	Visual table occupancy grid (Available, Occupied, Reserved) with sub-station filtering (Main Hall, Patio, Bar).
Inventory & Barcode	/products/scan/:code	GET POST	Dynamic SKU/barcode generation, barcode scanner lookup, thermal label printing metadata, and stock adjustments.
Menu Catalog	/menu-items	POST PATCH	Category management, dietary tags, price updates, and instant one-click stock availability toggle (86'ing items).
Voucher Engine	/vouchers	POST GET	Discount rule validation: enforces minimum order price, percentage/flat discounts, and one-time code usage checks.
Diagnostics Hub	/tickets/:id/messages	POST GET	Live support diagnostic thread: includes device ID, software build version, sync status, and messaging with Super Admin.

5 Relational Database Model (Prisma Schema Highlights)

The entity-relationship schema is optimized for ACID compliance, fast indexing on status and tenant fields, and cascading safety:

 Tenant & Identity Core	PostgreSQL	 Operations & POS Core	Indexed & Relational
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Business: id, name, email, phone, industry, subscriptionFee, allowedRoles, isActive.

User: id, name, email, password (bcrypt), role, pin (4-digit hashed), businessId (FK).

SubscriptionPlan: id, name, type, amount, currency, isActive.

Order: id, orderNumber, tableNumber, status, totalBill, notes, businessId (FK), tableId (FK).

OrderItem: id, orderId (FK), menuItemId (FK), name, quantity, unitPrice, totalPrice.

RestaurantTable: id, tableNumber, capacity, section, status (AVAILABLE/OCCUPIED/RESERVED).

6 Security, Performance & Resilience Engineering

 **Enterprise Tenant Scoping:** Every repository call executes an internal tenant validator (`getEffectiveBusinessId`). This guarantees that even if a malicious user crafts an ID belonging to a competitor restaurant, the Prisma ORM automatically blocks cross-tenant access.

Defense-in-Depth Security

- **Bcrypt Encryption:** Both user passwords and 4-digit quick POS access PINs are encrypted using salted bcrypt hashing.
- **Guard Layering:** Global `JwtAuthGuard` paired with custom `RolesGuard` on sensitive mutation controllers.
- **Data Validation:** Automated DTO validation via class-validator ensures sanitization before reaching business logic.

High-Concurrency Performance

- **Sub-Millisecond Queries:** Strategic indexing on `(businessId, status, barcode)` ensures rapid lookups.
- **Connection Pooling:** Managed PostgreSQL connection pool prevents memory exhaustion under heavy rush-hour orders.
- **Asynchronous Ticket Streaming:** KDS ticket bumping operates without blocking client checkout requests.

7 Business Impact & Quantifiable Results

Metric / Indicator	Legacy Operations	With Restaurant SaaS POS	Net Business Impact
Order-to-Kitchen Turnaround	8 to 14 minutes (paper slip)	< 30 seconds (instant KDS)	92% faster dispatch
Billing & Settlement Speed	3-5 mins per table checkout	15-20 seconds (MFS/Card grid)	3x faster table turns
Inventory Shrinkage	~8% - 12% untracked loss	< 0.5% with barcode logging	Massive margin protection
Multi-Branch Governance	Manual spreadsheet reconciliation	Real-time unified dashboard	100% automated auditing

8 Conclusion & Future Roadmap

The Restaurant POS & Management SaaS backend demonstrates how modern architectural patterns (NestJS, Prisma, PostgreSQL, Multi-Tenancy, and Granular RBAC) transform restaurant operations into a streamlined, error-free, and profitable enterprise. The platform serves as a blueprint for high-reliability F&B SaaS solutions.

Future Roadmap Horizons:

- **Offline-First Synchronization:** Service worker & IndexedDB local caching for uninterrupted offline billing during internet drops.
- **AI-Powered Menu Intelligence:** Automated demand forecasting and recipe cost-margin optimization.
- **QR Dine-In Table Ordering:** Direct customer ordering from mobile browser integrated with live KDS queue.

